



RESEARCH ARTICLE

Adapting funding processes for agile IT projects: an empirical investigation

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Abstract

Agile software development that provides software development organizations, the ability to respond to changes in turbulent business environments, has been gaining wide adoption. Agile software development projects are characterized by 'just enough' planning and lack of upfront commitment to scope, cost, and schedule. These characteristics pose conflicting demands on managers responsible for making funding decisions, because traditional approaches to funding IT projects are often based on well-defined scope, cost, and schedule. These conflicts demand the adaptation of traditional funding processes to suit to agile projects. We draw from Adaptive Structuration Theory to understand the nature of conflicts between traditional IT project funding processes and the dynamic nature of agile projects, and how these conflicts are addressed by practices that are appropriated in the process of social interaction between funding decision makers and development teams. On the basis of a multisite case study, we present a framework that explains how organizations adapt traditional IT funding approaches to accommodate the unique characteristics of agile IT projects.

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Introduction

The dynamic environment in which businesses operate today challenge the traditional way of software development and consequently the processes used for funding software projects. Software development organizations are becoming increasingly cognizant of the benefits of agile software development methods in such environments (Abrahamsson *et al*, 2003). For example, when regulatory requirements change frequently or when time-to-market constraints pressure organizations to quickly deploy IT solutions, organizations resort to agile methods. Agile methods are appropriate when project scope and requirements change rapidly throughout the project life cycle. Using short development cycles, 'just enough' planning and documentation, and working code, agile methods deliver value by their ability to adapt to changing requirements (Highsmith & Cockburn, 2001). Such tenets however conflict with the traditional process followed for IT investments. IT investments may be defined broadly to encompass all IT-related expenditures including people, training, documentation, consulting, external services, equipment, software, networks, and communications (Weill & Olson, 1989; Weill, 1991). For the purposes of this research, IT funding decision process involves making go/no-go and continuing funding decisions with the objective of maximizing benefits subject to constraints such as costs, schedule, scope, resources, and risks. While the literature on project management has long recognized the role of cost, scope, and schedule that are referred to as

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the triple constraints (Wallace & Keil, 2004), the importance of considering risks and resources as constraints has been recently recognized (PMI, 2008). Research on IT investments also recognizes that balancing risks and rewards is a commonly used approach to justify investments (Benaroch, 2002). The interdependencies among the constraints make IT funding decisions very challenging.

The use of traditional IT funding decision-making process in agile development presents conflicts due to inherent mismatches in their respective foci. Most traditional IT funding processes make the important assumption that the scope of the project is well-defined at the outset. They typically require an upfront estimation of the cost and schedule. However, this approach is often inconsistent with the requirements of agile development. Evolving requirements, rapidly changing technology, and time constraints cause planning and control for agile software development to be different (Cockburn, 2001; Levine *et al*, 2002). The conflicts between traditional IT funding process and agile methods demand the adaptation of the IT funding process to suit to agile projects. Motivated by this issue, our research seeks to answer the following research question: *How are IT funding decisions adapted for agile development projects?* This paper answers this question through a multisite case study of software development organizations involved in agile IT projects.

In addition to the literature on IT funding, we also draw from the literature on Adaptive Structuration Theory (AST). We characterize the funding process for agile development as a process of social interaction that involves appropriation of traditional process. AST is useful in developing insights on how traditional IT funding process is adapted to suit to agile development methods, thereby addressing the conflicts.

While our research question focuses on a practically relevant problem, it also helps us address a theoretical gap. AST has been largely used to study how advanced IT systems are appropriated in social interaction (DeSanctis & Poole, 1994). Our study examines how AST can be expanded to examine process adaptations, particularly when IT funding decisions are made in agile projects. Traditionally, AST has not been used to examine appropriation of structures under conditions that pose conflicting demands. Our study addresses this gap by studying IT funding process in agile development projects.

The paper is organized as follows: The next section presents the literature on IT funding and AST. We then present the research methodology and findings in the form of a theoretical framework. The discussion section relates our findings to AST and other theories. Finally, we present our contributions to theory and practice, and limitations and future research directions.

Theoretical background

IT funding process

A common approach to evaluating IT investments is the capital budgeting process, which relies on traditional

financial criteria (Bacon, 1992) and uses measures such as cost-benefit ratio, payback, return on investment (ROI), and net present value (Brealey & Myers, 1991; Emery & Finnerty, 1991; Bacon, 1992; Post *et al*, 1995; Apostolopoulos & Pramataris, 1997). These approaches are considered as objective and theoretically well-grounded (Dos Santos, 1991). To mitigate the limitations in using financial measures when evaluating IT funding decisions, other criteria have been proposed. For example, Parker & Benson (1988) identify six classes of value derived from IT: return on investment, strategic match, competitive advantage, management information support, competitive response, and strategic IS architecture. Bacon (1992) uses this framework to identify a set of criteria classified into three categories (financial, management, and development) that may be considered in IT investment decisions. IT investments may be grouped into five categories: strategic, informational, transactional, infrastructural, and threshold (Weill, 1991; Weill & Olson, 1989; Weill & Broadbent, 1998). These authors suggest that firms apply a contingency theory approach to decision making, where the type of funding and the context of the project determine the criteria used in funding decisions. In this research, we use costs, schedule, scope, resources, and risks as critical factors in evaluating IT projects.

Increasing empirical evidence suggests that project teams often adapt agile methods to suit their specific needs (Fitzgerald *et al*, 2006; Maruping *et al*, 2009; Vidgen & Xiaofeng, 2009). However, the requirements of agile development methods pose conflicts with the use of traditional funding processes. First, agile methods advocate moving into development without upfront definition of all the requirements, while traditional funding techniques that use financial measures (e.g., ROI) require a comprehensive understanding of the expected benefits (and hence the functionalities of the system) as well as costs. Second, project plans in agile development are adjusted frequently throughout the development process (Beck & Fowler, 2001), while the traditional funding decisions use project plans as benchmarks. Also, while traditional funding processes assume well defined project scope, agile methods may require changes to project scope to meet schedule constraints imposed by time-boxed development. While recent research recognizes that such differences require reexamination of traditional IT project funding processes (Thomas & Baker, 2008) and proposes alternative mechanisms such as incremental funding (Denne & Cleland-Huang, 2004), research on this important topic is quite nascent.

AST

AST attempts to explain how people incorporate advanced information technologies into their work processes (DeSanctis & Poole, 1994). AST posits that structures take two forms – those embedded in technologies/organizational processes and those embedded in social action. Understanding organizational changes that pertain to these structures involves gaining insights

into how they interact. AST presents two important concepts – structuration and appropriation. Structuration refers to the process of bringing rules, resources, and other structures into action, while appropriation refers to the application of these structures in a particular context. Four aspects of appropriations have been identified by past research (Billingsley, 1989; Sambamurthy, 1989; DeSanctis *et al*, 1991; Zigurs *et al*, 1991) – appropriation moves, faithfulness of appropriation, instrumental uses, and attitude. Appropriation moves refer to how groups adapt structures. Faithfulness of appropriation refers to the extent to which structures in use are consistent or faithful to their spirit. Instrumental use refers to the purpose of appropriation. Finally, attitude refers to the perceptions and feelings of users toward the structure.

Prior research in information systems has applied and extended AST in various ways. AST has been recognized as a useful meta-theory to study information systems research (Bostrom *et al*, 2009). A recent report on the 20-year program of interdisciplinary research on group support systems (GSS) highlights the importance of AST and its use in a wide variety of domains (DeSanctis *et al*, 2008). Key elements of AST – structuration and appropriation – have been extensively used to examine the use of GSS (e.g., Chidambaram *et al*, 1990; Gopal *et al*, 1992; Dennis & Garfield, 2003). To explain the effectiveness of global virtual teams, Maznevski & Chudoba (2000) examine the integration of structural and process elements by drawing a template of categories and subcategories from AST. Due to the increased use of AST in IS research, scales have also been developed for key AST constructs such as faithfulness of appropriation and consensus on appropriation (Chin *et al*, 1997; Salisbury *et al*, 2002). In analyzing the interplay between process variables and meeting outcomes and understanding the role of contributors of process losses, Reinig & Shin (2002) examine the validity of theoretical models that explain GSS use by drawing from the basic tenets of AST and other process-oriented theories. Drawing from AST, Niederman *et al* (2008) examine structuring tactics and develop a model that includes three levels of abstraction – meeting, activity, and real-time intervention. AST has been used to examine the role of facilitators in shaping group meeting outcomes (Anson & Bostrom, 1995). Study of creative processes and idea generation in GSS has found AST to provide a useful theoretical foundation in examining structures that shape these processes (Nagasundaram & Bostrom, 1994). AST has also been integrated with other models such as concerns-based adoption in examining adoption of ERP systems in academic settings (Lerouge & Webb, 2004) and diffusion theory to explain the adoption of electronic billing processes (Schwieger *et al*, 2004). AST has also been useful in examining how computer-mediated communication is shaped by organizational context – competitive or cooperative – and face-to-face interactions (Hill *et al*, 2009). This research observes that organizational context plays an important role in how participants overcome the

impact of computer-mediated communication on trust. Drawing from AST, Gupta & Bostrom (2009) develop a comprehensive model to examine technology-mediated learning. AST has also been used to examine the adoption of standardized systems by focusing on how structures influence adoption of systems (Arbin, 2008). Deviating from the conventional use of AST, Payton & Kiwanuka-Tondo (2009) adapt AST for the context of not-for-profit ICT use based on an action research study. While AST has traditionally been used to examine the appropriation of advanced information technologies, it has been adapted to study adaptation of methods as well. For example, AST has been extended to explain how agile methods are adapted based on project, organizational, and developmental contexts (Cao *et al*, 2009). Highlighting that prior use of AST has focused primarily on appropriation, Markus & Silver (2008) develop the concepts of structural features and spirit by introducing three new concepts – technical objects, functional affordances, and symbolic expressions. While technical objects comprise ‘IT artifacts and their component parts’, functional affordance ‘identifies what a user may be able to do with the [technical] object, given the user’s capabilities and roles’. Symbolic expressions refer to ‘communicative possibilities of a technical object for a specific user group’.

Although AST has been widely used to understand the role of social structure defined by rules and constraints of information technologies, it can also be used to examine structures from processes and organizations (DeSanctis & Poole, 1994). Drawing from the above literature, we argue that while AST has been used in a variety of contexts to study structuration and appropriation, it has not been used in contexts that pose conflicting demands on project teams’ needs. Such contexts present opportunities to extend AST. Here, we draw from AST to understand the structure imposed by organizational processes that suggests a particular set of criteria and procedures for IT funding decisions. We examine how appropriation is also shaped by the dynamics of the environment that demands the use of agile methods. The context presents conflicts between traditional IT funding criteria and requirements of agile projects. We use sources of structure, internal system, social interaction (which includes appropriation of structures and the process shaped by appropriation) as seed concepts in analyzing our data.

Research methodology

Our research is exploratory rather than confirmatory, in that we examine the phenomenon of funding in agile IT projects, which is a relatively uncharted area of research; hence our use of case study design (Yin, 2003). Since research in this area is nascent and lacks systematic frameworks that explain the dynamics involved in funding agile projects, we seek to inductively develop a framework using case study research (Eisenhardt, 1989). Qualitative methods facilitate investigation of complex phenomena and help develop richer insights.

Investments in agile projects have to be examined in their real-life context for us to develop an understanding of the boundary between funding processes, development methods, and the context in which the methods are used. The case-study method is suitable when the existing body of research in the area of focus is insufficient to develop causal questions requiring in-depth investigations (Paré, 2004). The funding process used in agile development projects is indeed such an area of study that requires further investigation.

Site selection

Theoretical sampling (Patton, 1990) was used in selecting one agile project each from 12 organizations. The following criteria were used in the selection: 1) each project used agile methods such as XP or SCRUM or adapted agile development approaches, 2) each project was recently funded, and 3) funding decisions in each project had faced challenges. Various team members from these projects who had rich positive and negative experiences pertaining to funding processes were identified as subjects for our study. Using project teams from various organizations helped us replicate our findings, thereby strengthening generalizability (Eisenhardt, 1989; Miles & Huberman, 1994; Yin, 2003). The 12 projects represent a rich mix of projects from health-care systems to desktop support tools. A summary of the characteristics of the study participants is presented in Table 1.

Data collection

Data were collected through multiple methods: semi-structured interviews, participant observations, and review of documentation. Within each organization, we interviewed a variety of stakeholders including top management, product managers, developers, senior architects, and project managers. We also interviewed customers or surrogate customers whenever possible. Data collection and analysis were conducted progressively, in an iterative fashion over a period of 18 months. We also used snowball sampling, in that we initially identified key contact persons (such as the project manager) in the case study sites, discussed our research objectives with them, and sought their help in identifying further participants. Each participant was interviewed typically two or three times to collect data and seek clarifications.

Our interviews centered on projects that faced conflicts between traditional funding process and the use of agile methods. We investigated various criteria that were used in making funding decisions, conflicts posed by these criteria and the tenets of agile development methods, and the practices that addressed these conflicts. Each interview lasted from 1 to 2 h. All interviews were audio recorded and transcribed. Detailed notes were also taken during the interviews. Requirement documents, project plans, process reports, meeting minutes, and project contracts were reviewed to enable triangulation. One author also participated in release planning meetings and iteration planning meetings of three projects

(DesktopSys, OnlineSecurity, EConnection). The meetings were recorded and meeting minutes were collected. The resulting release and iteration plans were also collected. Data collection was tightly interwoven with data analysis. Follow-up interviews were also conducted to gain further insights on the concepts that emerged from initial data analysis. As new data were collected, they were used to refine and/or modify concepts that were identified early on, and guide further data collection. Data collection and analysis were undertaken until 'theoretical saturation' (Eisenhardt, 1989) was reached, wherein further data did not provide any new insights.

Data analysis

We coded the interview data, initially to identify the following seed concepts drawn from our review of literature: sources of structure, conflicting demands, internal system, and social interaction process. New concepts that emerged from our analysis were then related to the seed concepts. These constructs related primarily to social interaction that shaped the adaptation of IT funding processes. Text fragments from the interview transcripts were tagged with one or more codes along with justifications for the selection of codes. Finally, key themes were identified based on the codes that emerged (Patton, 1990), which were then synthesized into a framework (presented in the next section). Two researchers separately examined each case for evidence. This evidence was summarized and entered into a matrix created for the purpose of intermediate qualitative data analysis (see Table 2). For each case, the authors recorded information about the source of structure (structure of funding process, other types of structure), conflicts between funding process and agile development, and social interactions (appropriation of structures, appropriated IT funding process). Each of the two authors analyzed the data independently and then combined their analyses by resolving discrepancies between each others entries through discussion, referencing back to the interview reports, and consulting with another author (inter-rater agreement- $\kappa = 0.94$).

The projects we studied also include a good mix of internal and external projects. There were no critical differences in our findings between these two types of projects. Any differences between internal and external projects that emerged in our analysis are presented in the findings. One reason for the lack of critical differences between the two types of projects is that even in internal projects that we studied, the organizations had also required the internal IT organization to provide documentation similar to that provided by external vendors.

The data analysis process described above helped us establish the plausibility and consistency of the concepts, subcategories, and relationships that emerged and ensure that our conceptual framework satisfies established criteria for the credibility and validity of qualitative research (Miles & Huberman, 1994; Pawlowski & Robey,

Table 1 Characteristics of study participants

<i>Project</i>	<i>Product description</i>	<i>Project information</i>	<i>Participants</i>
<i>DesktopSys</i>	Desktop support tools	Project duration: 9 months Team size: 5 people Method: XP External project	Project lead, senior developer
<i>EBank</i>	Banking system	Project duration: 36 months Team size: 22 people Method: XP External project	Project manager
<i>EConnection</i>	Packaged software for e-business connections and transactions	Project duration: 10 months Team size: 15 people Method: XP and SCRUM External project	Project manager
<i>EHealth</i>	Health-care system to manage health-care information flow	Project duration: 8 months Team size: 10 people Method: XP Internal project	Product manager (surrogate customer), project manager, senior software engineers
<i>EgovSys</i>	E-government application	Project duration: 6 months Team size: 10 people Method: XP and SCRUM External project	CIO, vice president, project manager, developer
<i>EPayment</i>	Online financial transactions support	Project duration: 8 months Team size: 6–9 people Method: XP and SCRUM External project	CIO, project manager
<i>MarketSys</i>	Marketing Support System	Project duration: 7 months Team size: 15 people Method: XP External project	General manager, project manager, project lead, developers
<i>NetSys</i>	Network system	Project duration: 8 months Team size: average 12 people Method: XP External project	General manager, senior software architect, client representative
<i>LogSys</i>	Internal application for logistics	Project duration: 8 months Team size: 12–15 people Method: XP Internal project	CIO, senior manager, project manager, project sponsor
<i>OnlineSecurity</i>	Internet security software	Project duration: 6 months Team size: 10 people Method: XP Internal project	Software engineer, project lead, product manager (surrogate customer)
<i>ShareSys</i>	Application that facilitates Inter-organizational data sharing	Project duration: 14 months Team size: 14 people Method: XP/SCRUM Internal project	President and CEO, VP technology operations director of marketing research, CIO, developers
<i>SupplySys</i>	Supply chain management system	Project duration: 10 months Team size: 14 people Method: XP Internal project	President, project manager, project lead, developers

2004). While no attempt is made to statistically evaluate the strength of concepts in the framework (Strauss & Corbin, 1990), theoretical relevance of concepts was established by their repeated presence or notable absence

(Strauss & Corbin, 1990). Findings from such analyses may be detailed and particularistic. However, we can use these findings to develop a general explanation (Eisenhardt, 1989; Orlikowski, 1993) through analytic

Table 2 Evidence included in the analysis matrix

Type of evidence	Information captured	Examples
Source of structure	Structure of process Other source of structure	Description of the traditional funding process Motivations for the adoption of agile methods Agile development practices adopted
Conflicts	Conflicts between traditional funding processes and agile projects	Challenges in reconciling evolving requirements in agile projects and upfront specification necessary for traditional funding processes
Social interaction	Appropriation of structures	Some traditional funding processes are adopted directly and some are adapted to suit the context of agile development
	Appropriated IT funding process	Concrete funding practices in agile development

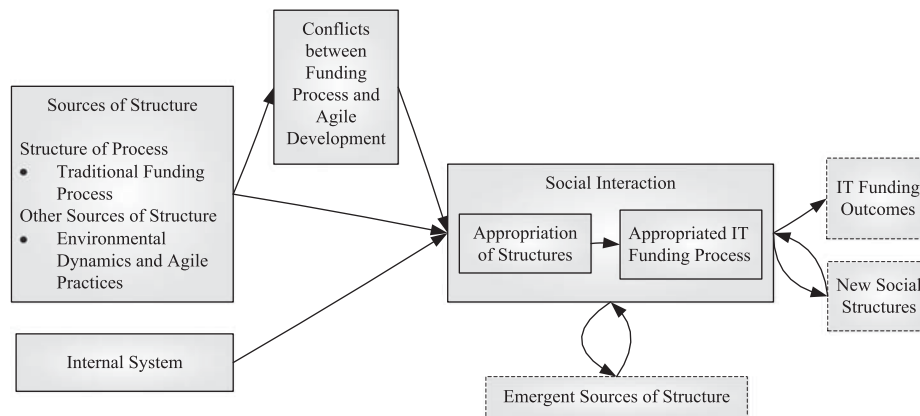


Figure 1 Conceptual framework.

generalization (Yin, 2003). Minor adjustments were made to the terminology used and the framework based on further feedback from research participants.

Findings

Our framework

Figure 1 depicts the framework that we have developed based on the analysis of our data. The AST framework (DeSanctis & Poole, 1994) was extended to include conflicts between funding process and agile development. We observe that the structure posed by traditional IT funding process shapes social interaction while also resulting in conflicts with agile methods. Such conflicts, in turn, shape social interaction. Our observations highlight that the conflicts are themselves shaped by the structure provided by traditional IT funding process and the need to use agile methods due to the dynamic nature of the environment. The sources of structure, such as traditional IT funding processes and the environmental dynamics and agile development result in conflicts, which then shape social interaction. Internal system that represents the nature of the project stakeholders and their relationships in the funding process also shapes social interaction. Appropriation of structures shapes the IT funding process. While we include emergent sources of

structure, IT funding outcomes and new social structures in the framework for the sake of consistency with AST (shown as dotted boxes in the framework), we restrict our discussion to the primary focus of our study – that is, conflicts and the process of social interaction that are presented in the following subsections.

Sources of structure

Traditional funding process as structure of process: Traditional funding process includes methods used in making funding decisions that consider both tangible and intangible benefits. Traditional funding process is based on the assumption that project scope is well-defined, upfront and fixed. Estimates of scope, cost, and schedule are used to evaluate the project.

Environmental dynamics and agile development as other sources of structure: Funding decisions are contingent on business environment and development methods. Dynamic environments faced by organizations dictate the use of agile development methods that constitute another source of structure. However, agile development challenges the funding process used in traditional plan-driven projects.

Internal system

The interaction between project stakeholders influences how the funding process is appropriated for agile projects. Our data shows that the relationship between the customer and the development team is the most critical factor in shaping funding decisions. Other factors such as the product sponsor's experience and knowledge of agile development also impact the adaptation of the funding process.

Conflicts between funding process and agile development

We present six conflicts between traditional funding processes and agile development that were identified in our study.

1. Upfront vs evolving functional requirements

Traditional funding decisions are made largely based on well-understood functionalities that the system will provide. However, agile development is most appropriate when the functionalities and the priorities assigned to them are evolving. Therefore, it is difficult to develop accurate estimates of the cost during the early stages of a project because agile development is characterized by unstable problem domains, requirement volatility, and dynamic planning and design phases. Furthermore, cost and schedule estimates are adjusted over time during the development process. A project manager at ShareSys recognizes these problems: *The project is initiated by a need recognized by the customer. However, even the customer does not fully understand the specific capabilities that can be developed. Also, since some [capabilities] are very novel, the customer does not have a full understanding of how they may evolve until some development is done.*

Since agile approaches do not have a formal requirement analysis phase, the initial estimation of project size is based on the known user stories. This estimate serves only as a starting point and is updated frequently. For example, for the OnlineSecurity project that involved 30 iterations, less than 50% of the tasks were created during the exploration and analysis phases, 25% new tasks were created during the next 10 iterations, and 25% were created in the last 10 iterations. This amounts to serious underestimation of the effort during early phases of development (by 100%).

2. Early vs delayed implementation of non-functional requirements

Traditional funding processes require that non-functional requirements are explicitly identified and considered during the design of the architecture. In agile development, the implementation of non-functional requirements is often delayed to later development cycles. Without detailed upfront design, estimating the cost and schedule associated with these requirements is challenging. Also, non-functional requirements are dependent on the functional requirements that are continuously evolving. In EgovSys, this issue was a

source of major concerns. *We know that performance and security considerations are very important for the customer, and they may also dramatically affect the cost of the project. But, we do not even fully understand the functionalities we are going to implement. Therefore, it is too early to worry about performance and security. Also, in this project, delivery speed is important. In early stages, the customer is willing to forgo some performance for speedy delivery.* The unknown degree of delay associated with the implementation of non-functional requirements posed difficulties in estimating and planning development effort. Also, the cost of implementing a non-functional requirement in agile development heavily depends on when they are implemented. *If we add it [non-functional requirement] early, it is easier [cheaper] to initiate, but we would spend effort to maintain it the rest of the project. Alternatively, we could delay pulling it until later, but it would be more expensive [to initiate] then.*

3. Specific completion date vs fluid delivery

Traditional funding processes usually require a specific delivery schedule for a project. However, the projection of the completion date for the project is discouraged in agile development because the focus on embracing changes makes such projections meaningless. Each release that lasts from 3 to 6 months is planned, but the completion date for the entire project itself is difficult to predict. As explained by one senior software engineering in EHealth: *Initially, we give estimate, if someone asks, on how long it will take to complete the project based on the body of work that we know of and understand. But the project changes its size and its requirements many times over the duration of the project. We cannot give a good estimate on things we don't know.* A project manager at SupplySys states: *Since the project scope itself was expected to evolve dramatically, it was difficult to provide a specific schedule for the project. The customer was willing to accept an open ended schedule for the project as long as we were delivering functionalities incrementally. Also, they [customer] will drive what will be delivered in each development cycle.*

4. Fixed vs flexible agreement

Traditional funding processes usually require an agreement between the sponsor and the development team to ensure the project satisfies user requirements. Project plans that define the scope, cost, and schedule explicitly are often used to ensure that both the sponsor and the development team have a shared understanding of the critical outcomes of the project. However, in agile development, it is common that neither the client nor the development team is able to completely articulate the scope of the project, and therefore neither party is in a position to finalize the cost, scope, or schedule for the project. It is very challenging to create a contractual agreement between the sponsor and the development team with evolving scope, cost, and schedules. *The client's accounting department wants to know how much the project will cost, and when they can expect to receive working*

software, states the project manager in NetSys. This concern was also highlighted in the difficulties faced by MarketSys while negotiating a contract with its customer. *The guiding philosophy in this project was – provide the customer whatever functionality they want. We recognized that this is too open-ended for us to precisely specify costs and schedules for the project in a contract. We were willing to work with the customer and assume the risk such an open contract exposes us to. Senior management made it clear that quickly delivering the functionality that the customer wants is the priority – at whatever cost.*

5. Plan-based vs continuous project control

Traditional funding processes use control mechanisms to lower project risks. For example, milestone reviews, documentation, and periodic reporting procedures are set up to monitor project progress. However, agile development emphasizes collaboration between customers and developers, rather than the use of phased milestones, heavy documentation, and formal reporting procedures. Instead, continuous control mechanisms such as feedback from short iterations, informal communication, and constant collaboration between customers and development teams are used. A developer at ShareSys states: *Rather than a mindset where we are measuring progress against a well defined plan that is typical in most of our projects, we are operating with a different mindset. We are here to develop the system closely working with the customer. Rather than specifying what they want upfront and monitoring whether the project is delivering these, the customer and the development team are working together to define the project as it is ongoing.*

6. Upfront design cost vs incremental design cost

Traditional funding processes consider design as a separate phase of development and estimate the cost associated with this phase based on factors such as project size and complexity. In contrast, agile development uses incremental design that is spread through the whole development process rather than completely upfront design phase. Specifically, refactoring is used to keep the quality of the design at an acceptable level. Refactoring costs depend on the extent to which development practices such as unit testing and pair programming are used. However, the costs associated with these practices are not given sufficient consideration in the estimation of project costs. The project manager for OnlineSecurity explains the issue: *Refactoring takes a large portion of development effort. I will say 30% but I am not sure as it is difficult to separate refactoring from other development. However we are not paid to do it. We are not paid to write unit tests either. They are overhead. The customer doesn't care if we refactor our code.*

Social interaction

The sources of structure, internal system, and the conflicts described above shape the process of social interaction involved in funding decision making in agile

projects. The following section describes the various appropriated practices that were observed in our study, how these practices address the conflicts described above, and the implications of these practices for the funding process. The practices described below demonstrate that the funding process, tailored for agile development, involves social interaction characterized by continuous collaboration and negotiation among the stakeholders involved.

Appropriation of structures The process of appropriation of structures includes appropriation moves, faithfulness, and attitude toward funding agile projects. A variety of appropriation moves were used by project teams to address the challenges described above. For example, one type of move was to directly use the existing structures at the beginning of the project and then adapt them as the project unfolds. The focal projects challenged the basic assumption behind traditional funding processes that project scope can be well-defined upfront. For example, EgovSys used a traditional funding project at the beginning assuming a fixed scope for the project. This appropriation was faithful to the traditional funding process as the scope was pre-determined. However, the project changed to a 'fixed price variable scope' funding mechanism later, which is less faithful to the spirit of traditional funding processes but more faithful to the spirit of agile development. Some features from existing IT funding process were adapted for instrumental use. For example, flexible funding processes were used to negotiate agreements between the product sponsors and development teams so that the risks were controlled and mitigated for both parties. Product sponsors and the developers have different attitudes toward the appropriated funding processes. Product sponsors prefer an approach that focuses on outcomes, whereas developers prefer a contract that can accommodate changes in scope and hence process.

Appropriated IT funding process We identified eight practices that address the conflicts posed by agile development and traditional IT funding processes.

1. Negotiable project scope

The project stakeholders explicitly recognize that the scope of the project will be negotiated between the customer and the development team continuously as the project progresses. This practice applies for projects that have a fixed completion date but not a fixed scope. An important task in agile planning is the adjustment of project scope so that the software may be released on time. For example, as the requirements of DesktopSys in the contract were defined at a high-level, the project team categorized the requirements into two groups: above the line and below the line requirements. Above the line requirements were delivered under this contract, and below the line requirements would only be incorporated into the current version if there were sufficient

time. Otherwise, they would be captured as requirements change and considered for inclusion in the next version. While the broad requirements specified in the contract were not negotiable, user requirements that were developed based on these requirements may be incorporated into different versions of the product as permitted by the product development cycles.

The ability to negotiate which non-functional requirements will be implemented and when they will be implemented provides much needed flexibility with funding decisions. For example, in ShareSys, *the customer decided during each iteration whether it is necessary to focus on adding more functionality or improving the existing functionality by focusing on concerns like security and response time. We [development team] provided the customer with our estimates of the effort needed and the schedule for each of the various options on the table.* Cost estimates for implementing non-functional requirements were adjusted during iterations. As the project manager at EPayment explains: *We keep all non-functional requirements on a wiki and reference that requirement in each story where it is applicable. In the same time, we write these requirements up as stories and are scheduled into the iterations. This effort [time] are counted in estimation.*

This practice addresses the conflicts 'upfront vs evolving functional requirements' and 'early vs delayed implementation of non-functional requirements'. First, for fixed schedule projects, some initially planned features are dropped if the product cannot be delivered on time. Second, the ability to negotiate project scope allows the delayed estimation and implementation of non-functional requirements. As explained by a project manager in LogSys: *Non-functional requirements are identified during the development and are managed because of the flexibility [of project scope]. Customers were provided with the [estimation] information and they made implementation decisions.*

2. Visibility of agile process

Visibility is important in funding agile projects, as elaborated by the client of NetSys: *Our contracting officer requires a list of functionalities and will check at the end of the project to confirm that the vendor has delivered them. The list can be updated but the changes need to be reported [to contracting offer]. For agile projects, we have to pay extra attention to make sure that happens.* Agile development provides visibility of development process, and therefore project stakeholders are able to consider the cost of incremental design in their funding decisions. Since customers are intensely involved in agile development, they are able to development a significant understanding of the impact of their requirements on costs and schedules. For example, *The customer was able to clearly see that meeting their performance criteria, response time, took such a lot of effort. In the subsequent cycles, they were a lot more careful with their requests and had a clear understanding of the effects on costs and schedule,* explained a developer at EgovSys.

This practice addresses the conflict 'upfront design cost vs incremental design cost'. Short development cycles and intensive customer involvement render the development process very transparent to customers. *Progress on the project is visible,* a project manager at EgovSys states: *This [visibility] was one of the major benefits of the process. We now have a customer who is much better equipped to understand and approve our cost and schedule requests.*

3. Contract risk management

Different types of contractual arrangements between customers and development teams are adapted while funding agile projects. Since embracing change is an important tenet of agile development, fixed price, fixed scope contracts that are common in traditional development are not the most appropriate vehicles in these projects.

However, some organizations still used these contracts for external projects because senior management was not willing to *assume the risks associated with an open ended contract,* says a project manager at EgovSys. *We agreed to work with such an arrangement even with the expectation that requirements may change during development. We were confident that a successful customer experience with the project will help create the confidence in more flexible arrangements in future projects. Even though this project was supposed to have fixed scope, we viewed it as a fixed price, variable scope contract and we assumed all the risks.*

A variation of this arrangement that was used in an internal project at SupplySys can be called a 'fixed price and negotiated scope' contract. Senior managers had set aside a fixed amount that will be invested in the project, which was charted to provide *ad-hoc* reporting and querying capabilities. *Since we were closely working with them [senior managers/customers] to understand their requirements and were providing cost and schedule estimates for each of the requirements, they [customers] could easily prioritize what must be implemented in each cycle and in the project.*

A 'pay as you go' model was used in another internal project, ShareSys, where the project was treated as several mini projects each of which developed a small set of features. Costs and schedules were agreed upon for each of the mini projects that lasted from four to six 2-week development cycles. *As we were dealing with small chunks of functionality, we funded each phase as it was implemented,* explains a senior manager.

This practice addresses the conflict 'fixed vs flexible agreement'. The risks are shared by both the client and the developers. A project manager at ShareSys explains: *This model [pay as you go] worked as the customer knew that there would be another release next phase and they were more willing to accept moving a feature to next phase. Customer liked this model as their risk was minimized to just one phase. On the other side, we worked hard to make sure the funding would be approved for the next phase.*

4. Trust building between the product owner (customer) and product developer

All the focal projects in the study emphasized the criticality of building trust using two different approaches:

(a) Start small

Instead of a large contract, the team starts with a smaller version of the project (fewer number of iterations). If the customer is satisfied with the delivered product and the development process, they are likely to expand the contract to add more iterations. In ShareSys, *the project was funded in small increments and the customer was working closely with the [development] team. It helped us avoid creating a lot of overhead like formal agreements and controls and focus on delivering the system features to the customer,* explains a project lead in ShareSys.

(b) Reduced risk for both the developer and the customer

The goal is to treat the customer as a business partner and foster a trusting relationship. In the arrangement between MarketSys and its customer *We had a clear mandate to please the customer and provide whatever the customer wants. Success in this project was considered important to develop a long term business partnership with the client. Therefore, we worked extra hard to make sure that every change that was desired by the client was accommodated. This process also helped them [customer] see that we were genuinely interested in the success of the project and it definitely increased their confidence in us,* elaborates a senior executive at MarketSys.

This practice addresses the conflict 'fixed vs flexible agreement'. The trust built between the customer and the developer helps customers accept an open-ended agreement. The customer of NetSys described: *[We chose flexible contract] because we had a lot of confidence in the vendor we had selected – If our vendor had not been so good, we would have had a few more reservations about entering into a contract like this.*

5. Just enough control

The focal projects adopted control mechanisms that were not overly burdensome, but provided enough control over the project to satisfy organizational norms for funding IT projects. While some of the documentation was still necessary, heavy documentation, process reports, and reviews were replaced by extensive communication between the team and the customers. For example, *as the executives who are interested in the project outcomes are closely working with the development team, we didn't want to impose burdensome documentation and reviews. These are required in the first place to make sure that the project is on track and for taking corrective action. In fact, with the close involvement, we can catch and correct problems well before we can do so in other projects. We don't have to wait for a review to figure out that the project is not performing well, and is not delivering what the users want. We can identify these problems much earlier,* explains a project manager at SupplySys.

This practice addresses the conflict 'plan-based vs continuous project control'. The balance between formal and informal controls satisfied the customer needs and facilitated agile development. *Though it looks like we have fewer checks and balances in the formal sense, we have better control though it is not very bureaucratic,* states the project manager at SupplySys.

6. Layered project scheduling

A balanced scheduling practice is adopted while funding agile projects. Traditional project schedules are replaced by layers of plans that include a release plan and a series of iteration plans. A release plan may cover multiple releases. *In some sense, we are breaking a project into multiple mini projects. When the time frame is short, our ability to provide accurate schedules is dramatically better,* says a project lead at ShareSys. *We need to know what we will get in each release and how long it will take to provide the release. Rather than waiting several months for the project to provide deliverables, the project team is able to provide us estimates of when important features will be provided in the various releases,* states a project sponsor (customer) at LogSys.

This practice addresses the conflict 'specific completion date vs fluid delivery'. Project completion date is decided by ongoing fixed-time releases. *The flexibility to break a project in multiple releases and each release into multiple iterations help us come up with schedules that we can meet with some confidence,* states the project manager at SupplySys.

7. Agile budgeting

Agile budgeting shifts the focus to continuously providing business value in contrast to traditional approaches that do not support the realignment of project priorities with changing business needs. In agile projects, both the customer and the development team take the view that the primary role of an IT project is to add value to the customer. The project sponsor (customer) of LogSys explains: *The development approach we are using helps us explore the requirements and needs of the customer as we are developing the system ... We are starting with some high level understanding of the capabilities that can be developed with new technology, but we don't have all the details of costs, timelines and benefits that are used in creating a business case in our company.*

In NetSys project, an initial 3-month startup period is used to organize the development teams and the governance structure. Subsequent to the startup period, funding decisions are made and reviewed every quarter. *Because we are working closely with the customer who is sponsoring the project, they can specify the immediate scope of the project over the next five or six development cycles – in the next quarter – and then provide funding for just these activities,* states a project manager at NetSys. Incrementally, funding the project as it is evolving gives the customer the ability to change the nature of the project at any time and support only the activities that delivers them value, states a project lead at SupplySys.

This practice addresses the conflict ‘plan-based vs continuous project control’. Regular project reviews through the development cycle provide effective control. These reviews ensure that value is delivered to the customer. The general manager at NetSys stated: *We focus on getting working software that delivers value to the users. Working software creates satisfied customers because they see what they get for their money.*

8. Agile estimation

The focal projects have adapted approaches to estimate both project scope and completion date. To estimate project scope at the beginning of a project, just enough requirements gathering and architectural modeling are conducted. The modeling effort provides just enough understanding to develop an initial budget and schedule without investing in excessive documentation. *We want to get started with the project with a small set of important requirements. We can refine these later as we begin to fully understand the capabilities that can be provided by the technology. We can provide seed funding for the project to get it started based on the estimated scope of the critical requirements,* states a senior manager in SupportSys.

A commonly adopted strategy for developing initial budget and schedule estimates involves seeking inputs from project personnel who have relevant experiences from similar projects. This initial planning process is carried out at a coarse level. Known functionalities are broken into tasks and the cost and schedule for each task is estimated. Many of these tasks may be discarded and many more may get added during development. EConnect uses a version of this strategy: *We estimate the size [of stories] by three different categories: known stories, stories without enough information and unknown stories. We add different buffer to each category. For unknown stories, we add 100% and for known [stories], only 10–20%.* A project manager in OnlineSecurity describes the process as follows: *So during the early iterations I add 50% to any estimates whenever anyone is computing a project completion date. Once we have a few iterations under our belt and our*

velocity is stable, I drop the multiplier down to 25%. Thus, initial estimates are typically provided with considerable buffers (30–50% on average) and these are continuously adjusted based on status changes.

This practice addresses the conflict ‘upfront vs evolving functional requirements’. Project cost is estimated by experienced engineers based on their understanding of high-level requirements. A developer at OnlineSecurity explained: *Requirements change ... we consider this when we estimate the project cost so it would not be a surprise later.*

Table 3 maps the specific practices that comprise the appropriated IT funding process described above to conflicts addressed by each practice.

Discussion

This study addresses our research question by developing a framework (Figure 1) that describes the adaptation of traditional IT funding practices for agile development. Our framework describes the role of conflicts between traditional IT funding process and agile development in shaping appropriation of funding processes. Past research on software projects in uncertain environments has considered process flexibility as an internal capability that needs to be adapted to suit changing customer needs. Such a capability may be developed through the use of agile development methods. This paper considers IT funding process and software development process as critical internal capabilities that are in need of mutual alignment.

Our findings highlight various adaptations to the IT funding process that have emerged as a process of social interaction among stakeholders. These stakeholders represent different business units that are responsible for budgeting for IT projects and the IT project team members who are responsible for making development-oriented decisions. Drawing from AST, our framework highlights that the structure provided by conventional IT funding processes conflict with the need to use agile methods in dynamic environments. This structure is then appropriated to make it suitable for the context at hand.

Table 3 Appropriation practices addressing conflicts

Appropriation IT funding practice	Conflict addressed	Number of projects that used the practice
Negotiable project scope	Upfront vs evolving functional requirements	5
Trust-building between the product owner (customer) and product developer	- Early vs delayed implementation of non-functional requirements - Fixed vs flexible contract	9
Visibility of agile process	Upfront design cost vs incremental design cost	10
Contract risk management	Fixed vs flexible contract	6
Just enough control	Plan-based vs continuous project control	7
Layered project scheduling	Specific completion date vs fluid delivery	11
Agile budgeting	Plan-based vs continuous project control	5
Agile estimation	Upfront vs evolving functional requirements	10

A set of appropriated practices then emerge in social interaction.

The use of AST in examining the funding processes highlights the importance of social interaction and the structures that shape this interaction. Also, AST has been typically used to examine organizational change shaped by the adoption of advanced information technologies such as Group Decision Support Systems (DeSanctis *et al*, 2008). In our study, the central artifact is the IT funding process. There are key differences that pertain to the use of AST to study a process rather than a technology. For example, the notion of restrictiveness, which refers to the degree of freedom a user has in applying technology, is not as critical in our context due to the inherent flexibilities of processes enacted by individuals when compared with what is afforded by information technologies that implement processes. Past research on AST has argued that 'spirit, features, and dimensions of a structure constitute the structural potential' (Bostrom *et al*, 2009). In our context, the different elements of the IT funding process, the assumptions behind it, environmental characteristics, and the characteristics of conflicts provide the structural potential, defining the characteristics of structure that is afforded and can be appropriated by the actors. Our findings also relate to functional affordances (Markus & Silver, 2008) advanced by recent research on AST in that the central artifact of focus affords goal-oriented action of stakeholders involved by setting boundaries that need to be considered in addressing conflicts. For example, our findings highlight that the IT funding process – the artifact under consideration here – affords the stakeholders the possibility for goal-oriented action in that it facilitates adaptations based on changing environmental conditions that demand the use of agile methods.

The conflicts that we have identified are similar in spirit to the paradoxes highlighted by past research (Lewis, 2000) that may stem from mixed messages and system contradictions. Mixed messages may result from inconsistencies between prescribed organizational practices and practices that emerge in action. For example, organization-wide policies on IT funding may be inconsistent with the need for agile development dictated by environmental conditions. This may, in turn, result in system contradictions that entail in organizational systems and processes that may not materialize in practice. Such environments that face tensions between control and flexibility have been characterized as social spaces with members pulling away in directions that may contradict one another (Lewis, 2000). Conflicting demands in such cases are often addressable by considering them in a broader context such as the organization, market environment, and strategy. For example, organizational stakeholders who may have conflicting views about investment decision making and development processes may be able to come together to a common understanding when these conflicts are placed in the context of the overall organizational goals. This can then

be complemented with organizational mechanisms that will foster an environment suitable for addressing conflicts. Our findings are consistent with this notion in that agile budgeting and estimation requires a good understanding of the rationale for such an approach driven by recognizing market conditions.

Organizational mechanisms such as meta-routines, job enrichment, switching, and partitioning have been identified by past research as enablers of the ability to address conflicting demands. While Adler *et al* (1999) argue that meta-routines attempt to address the conflicts by converting non-routine to routine tasks, our findings suggest a broader implication for the role of meta-routines. Due to the rate and extent of changes in agile environments, establishing meta-routines that encourage context-based adaptation of conventional processes emerge as viable solutions as opposed to routinizing non-routine tasks that have been developed to adapt to dynamic conditions. The practices that were observed in our study can be abstracted to develop meta-routines that enable stakeholders to simultaneously recognize the routine elements of IT funding and the role of agile methods in adapting to dynamic environments. While sequential switching and partitioning (Adler *et al*, 1999) might not be suitable in the context of our study, meta-routines should enable the development of an environment that fosters adequate autonomy to be able to focus on different organizational priorities based on contextual needs. Such adaptations are typically challenged by organizational inertia as routines become ingrained and gain significant fit with internal conditions. This might usually require organizations to completely reconfigure capabilities to suit to changing external conditions. However, in some cases, internal capabilities might have a residual fit with parts of requirements driven by external environment (Gilbert, 2006). This is evidenced in our study wherein elements of traditional IT funding process continues to have some residual fit while adaptations evolve to suit the need for agility. Gilbert (2006) also argues that cognitive framing of external changes as threats might motivate suitable response while that are framed as opportunities might lead to flexible plans and entrepreneurial growth. Our findings are consistent with this in that project teams tend to present the need for adaptable processes such as agile methods in terms of both the challenges shaped by rapid changes in environments and the opportunities for addressing evolving requirements. We observed that continuous negotiations along the triple constraints of scope, schedule, and cost are motivated partially by threats of existing processes becoming liabilities and by the opportunities to adapt to changing customer need. While Gilbert (2006) argues that the latter – that is, cognitive framing of external changes as opportunities – might not elicit strong commitment, we observed that commitment was developed over the life cycle through attempts at continuous alignment of process with customer

needs and the development of close working relationships with all stakeholders involved as advocated by agile methods.

Our study also extends the notion of behavioral integration of top management team that was examined by Lubatkin *et al* (2006). Lubatkin argues that this construct refers to senior management team's 'wholeness and unity of effort', leading to better synchronized social and task processes. Such teams are also better able to control the quality of information, collaborative behavior, and joint decision making (Lubatkin *et al*, 2006). Our findings suggest that behavioral integration of top management teams both from business units and from IT development, and across the development team leads to an emergence of effective and productive negotiations in social interaction characterized by appropriated practices. This is highlighted by various practices that were observed. For example, 'visibility of agile practices', retaining 'just enough control', and 'negotiable project scope' present a common thread of practices that indicates a better synchronized top management team that involves clients/stakeholders who need systems that solve problems in their business units and project executives that focus on putting in place mechanisms that deliver value to the clients. In this context, a behaviorally integrated top management team understands the implications of introducing agile methods in shaping IT funding decision-making processes. Development teams choose adaptable processes and rationalize them to IT investment decision makers, thereby helping shape an environment that addresses the conflicts between traditional IT funding processes and agile development.

Conclusions

Contributions to theory

Our study contributes to two areas of theory – adaptive structuration and IT funding.

Our study extends AST by incorporating conflicting demands that result from competing sources of structure. Also, appropriated practices that emerge in social interaction provide further insights into the process of appropriation of structures. While the notion of implementing meta-routines might further streamline the appropriation of these practices, our discussion of findings advocates for meta-routines that facilitate continuously evolving set of practices that adapt to contextual conditions. For example, organizations can institutionalize flexible contracting approaches that are context-dependent rather than top-down definition of one-size-fits-all contracts. This highlights the criticality of considering appropriation cycles that are continuously tweaked to align with changing environmental conditions.

In answering the question: *How are IT funding decisions adapted for agile development projects?* our study addresses a gap in the literature on IT funding process by highlighting the need to adapt traditional processes for

making funding decisions so that agile projects can be rationalized better. IT funding decisions that rely on significant upfront commitments to the classical triple constraints of cost, scope, and schedule may be unsuitable for agile projects that pose challenges in estimating these parameters. Our findings present a set of practices that balance the need for upfront definition of key parameters and continuous commitment to addressing changing customer needs.

Contributions to practice

Our study contributes to practice by providing key implications for stakeholders such as business unit managers who are involved in IT budgeting and IT project managers and developers who are involved in negotiating with business unit managers to define cost, scope, and schedule constraints.

Stakeholders involved in making 'go/no-go' decisions on IT funding and decisions on various funding parameters such as funding cycles and milestones should recognize the need for adapting their approach to agile environments. Stakeholders should collaboratively develop an appropriated funding process that is suitable for agile projects. Funding decisions should be based on continuous feedback from project team members and negotiations based on changing customer values. Stakeholders involved in IT funding and project development should recognize the pitfalls in applying traditional funding methods to agile projects. For example, our findings suggest that contract risk management practiced traditionally is likely to ignore the fact that the incorporation of agile methods makes it difficult to validate assumptions that are used in developing contracts. Force-fitting traditional approaches to contract risk management will thus result in irreconcilable conflicts between decision makers who approve funding for IT projects and project managers who select appropriate methodologies to implement such projects. Approaches that involve developing contracts with fixed price, but negotiated scope, and pay-as-you-go models have been identified in the focal projects we studied as solutions that address this conflict. Stakeholders who make IT funding decisions should recognize that the inability of project managers to present complete details of costs, timelines, and benefits of projects upfront is attributed to the nature of the development method used. To address this, projects can be funded in frequent cycles to match those of the agile method used, such that funding is approved for a limited, agreed upon scope, and revisited every few cycles. Such incremental funding provides benefits for both clients and the development teams, in that clients may have the ability to adjust the scope of the project in upcoming cycles while the development teams to provide solutions that are valued by clients.

Development team members who have embraced the tenets of agile methods have to recognize the need to be able to rationalize their approach to top management and product sponsors and participate in

negotiations to define funding cycles and the triple constraints. Stringent adherence to agile method without adequate attention to addressing the need for budget justifications might result in tensions that are not conducive to project success.

Limitations or future research

Generalizing our findings to other contexts that involve conflicting demands should be done with caution as the findings may be particularistic to use of specific tenets from agile and traditional approaches, and to the sites we

studied. Our study provides fertile ground for large-scale quantitative studies that can examine specific factors that help organizations in balancing conflict between agile and traditional approaches for IT funding. The link between ability to balance such conflicting demands and organizational performance is a subject of ongoing research. Future research can also focus on key differences between agile-traditional balance and other types of conflicting demands identified by the literature to gain insights into any differences in approaches to address such conflicting demands.

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References

- ABRAHAMSSON P, WARSTA J, SIPONEN MT and RONKAINEN J (2003) New directions on agile methods: a comparative analysis. In *Proceedings of the 25th International Conference on Software Engineering*, pp 244–254, IEEE Computer Society Portland, Oregon.
- ADLER PS, GOLDOFTAS B and LEVINE DI (1999) Flexibility versus efficiency? A case study of model changeovers in the Toyota production system. *Organization Science* **10**(1), 43–68.
- ANSON R and BOSTROM R (1995) An experiment assessing group support system and facilitator effects on meeting outcomes. *Management Science* **41**(2), 189–208.
- APOSTOLOPOULOS TK and PRAMATARIS KC (1997) Information technology investment evaluation: investments in telecommunication infrastructure. *International Journal of Information Management* **17**(4), 287–296.
- ARBIN K (2008) The structure of determinants of individual adoption and use of e-ordering systems. *Human Systems Management* **27**(2), 143–159.
- BACON CJ (1992) The use of decision criteria in selecting information system. *MIS Quarterly* **16**(3), 335–354.
- BECK K and FOWLER M (2001) *Planning Extreme Programming*. Addison Wesley Longman, New York, NY.
- BENAROCHE M (2002) Managing information technology investment risk: a real options perspective. *Journal of Management Information Systems* **19**(2), 43–84.
- BILLINGSLEY J (1989) An analysis of small group appropriation of decision structures in a computer-assisted group decision support environment. In *International Communication Association Annual Meeting*. San Francisco, CA.
- BOSTROM RP, GUPTA S and THOMAS D (2009) A meta-theory for understanding information systems within sociotechnical systems. *Journal of Management Information Systems* **26**(1), 17–47.
- BREALEY RA and MYERS SC (1991) *Principles of Corporate Finance*. McGraw-Hill, Inc., New York.
- CAO L, MOHAN K, PENG X and RAMESH B (2009) A framework for adapting agile development methodologies. *European Journal of Information Systems* **18**(4), 332–343.
- CHIDAMBARAM L, BOSTROM RP and WYNNE BE (1990) A longitudinal study of the impact of group decision support systems on group development. *Journal of Management Information Systems* **7**(3), 7–25.
- CHIN WW, GOPAL A and SALISBURY WD (1997) Advancing the theory of adaptive structuration: the development of a scale to measure faithfulness of appropriation. *Information Systems Research* **8**(4), 342–367.
- COCKBURN A (2001) Agile software development. In *The Agile Software Development Series* (COCKBURN A and HIGHSMITH J, Eds), Addison Wesley Longman, Massachusetts.
- DENNE M and CLELAND-HUANG J (2004) The incremental funding method: data-driven software development. *IEEE Software* **21**(3), 39–47.
- DENNIS AR and GARFIELD MJ (2003) The adoption and use of GSS in project teams: toward more participative processes and outcomes. *MIS Quarterly* **27**(2), 289–323.
- DESANCTIS G and POOLE MS (1994) Capturing the complexity in advanced technology use: adaptive structuration theory. *Organization Science* **5**(2), 121–147.
- DESANCTIS G, POOLE MS, LEWIS H and DESHARNAIS G (1991) Using computing in quality team meetings: initial observations from the

- IRS – Minnesota project. *Journal of Management Information Systems* **8**(3), 7–26.
- DESANTIS G, POOLE MS, ZIGURS I, DESHARNAIS G, D'ONOFRIO M, GALLUPE B, HOLMES M, JACKSON B, JACKSON M, LEWIS H, IMAYEM M, LEE-PARTRIDGE J, NIEDERMAN F, SAMBAMURTHY V, VICIAN C, WATSON R, BILLINGSLEY J, KIRSCH L, LIND R and SHANNON D (2008) The Minnesota GDSS research project: group support systems, group processes, and outcomes. *Journal of the Association for Information Systems* **9**(10), 551–608.
- DOS SANTOS BL (1991) Justifying investments in new information technologies. *Journal of Management Information Systems* **7**(4), 71–89.
- EISENHARDT KM (1989) Building theories from case study research. *Academy of Management Review* **14**(4), 532–550.
- EMERY DR and FINNERTY JD (1991) *Principles of Finance with Corporate Applications*. West Publishing Company, St. Paul, MN.
- FITZGERALD B, HARTNETT G and CONBOY K (2006) Customising agile methods to software practices at intel shannon. *European Journal of Information Systems* **15**(2), 200–213.
- GILBERT CG (2006) Change in the presence of residual fit: can competing frames coexist? *Organization Science* **17**(1), 150–167.
- GOPAL A, BOSTROM RP and CHIN WW (1992) Applying adaptive structuration theory to investigate the process of group support systems use. *Journal of Management Information Systems* **9**(3), 45–69.
- GUPTA S and BOSTROM RP (2009) Technology-mediated learning: a comprehensive theoretical model. *Journal of the Association for Information Systems* **10**(9), 686–714.
- HIGHSMITH J and COCKBURN A (2001) Agile software development: the business of innovation. *IEEE Computer* **34**(9), 120–122.
- HILL NS, BARTOL KM, TESLUK PE and LANGA GA (2009) Organizational context and face-to-face interaction: influences on the development of trust and collaborative behaviors in computer-mediated groups. *Organizational Behavior & Human Decision Processes* **108**(2), 187–201.
- LEROUGE C and WEBB HW (2004) Appropriating enterprise resource planning systems in colleges of business: extending adaptive structuration theory for testability. *Journal of Information Systems Education* **15**(3), 315–326.
- LEVINE L, BASKERVILLE R, LOVELAND LINK JL, PRIES-HEJE J, RAMESH B and SLAUGHTER S (2002) *Discovery Colloquium: Quality Software Development @ Internet Speed*. Software Engineering Institute, Pittsburgh, PA.
- LEWIS MW (2000) Exploring paradox: toward a more comprehensive guide. *Academy of Management Review* **25**(4), 760–777.
- LUBATKIN MH, SIMSEK Z, YAN L and VEIGA JF (2006) Ambidexterity and performance in small- to medium-sized firms: the pivotal role of top management team behavioral integration. *Journal of Management* **32**(5), 646–672.
- MARKUS ML and SILVER MS (2008) A foundation for the study of it effects: a new look at Desantis and Poole's concepts of structural features and spirit. *Journal of the Association for Information Systems* **9**(10), 609–632.
- MARUPING LM, XIAOJUN Z and VENKATESH V (2009) Role of collective ownership and coding standards in coordinating expertise in software project teams. *European Journal of Information Systems* **18**(4), 355–371.
- MAZNEVSKI ML and CHUDOKA KM (2000) Bridging space over time: global virtual team dynamics and effectiveness. *Organization Science* **11**(5), 473–492.
- MILES MB and HUBERMAN AM (1994) *Qualitative Data Analysis*. Sage Publications, Thousand Oaks, CA.
- NAGASUNDARAM M and BOSTROM RP (1994) The structuring of creative processes using gss: a framework for research. *Journal of Management Information Systems* **11**(3), 87–114.
- NIEDERMAN F, BRIGGS RO, DE VREEDE G-J and KOLFSCHOTEN GL (2008) Extending the contextual and organizational elements of adaptive structuration theory in gss research. *Journal of the Association for Information Systems* **9**(10), 633–652.
- ORLIKOWSKI WJ (1993) Case tools as organizational change: investigating incremental and radical changes in systems development. *MIS Quarterly* **17**(3), 309–340.
- PARÉ G (2004) Investigating information systems with positivist case study research. *Communications of AIS* **13**, 233–264.
- PARKER MM and BENSON RJ (1988) *Information Economics: Linking Business Performance to Information Technology*. Prentice Hall, Englewood Cliffs, NJ.
- PATTON MP (1990) *Qualitative Evaluation and Research Methods*. Sage, London.
- PAWLOWSKI S and ROBEY D (2004) Bridging user organizations: knowledge brokering and the work of information technology professionals. *MIS Quarterly* **28**(4), 645–645.
- PAYTON FC and KIWANUKA-TONDO J (2009) Contemplating public policy in hiv/aids online content, then where is the technology spirit?. *European Journal of Information Systems* **18**(3), 192–204.
- PROJECT MANAGEMENT INSTITUTE (PMI) (2008) *A Guide to the Project Management Body of Knowledge* (4th edn), *PMBOK Guide*[®]. Project Management Institute.
- POST GV, KAGAN A and LAU K-N (1995) A modeling approach to evaluating strategic uses of information technology. *Journal of Management Information Systems* **12**(2), 161–187.
- REINIG BA and SHIN B (2002) The dynamic effects of group support systems on group meetings. *Journal of Management Information Systems* **19**(2), 303–325.
- SALISBURY WD, CHIN WW, GOPAL A and NEWSTED PR (2002) Research report: better theory through measurement – developing a scale to capture consensus on appropriation. *Information Systems Research* **13**(1), 91–103.
- SAMBAMURTHY V (1989) Supporting group performance during stakeholder analysis: The effects of alternative computer-based designs. p 314, Doctoral Dissertation, University Of Minnesota.
- SCHWIEGER D, MELCHER A, RANGANATHAN C and WEN HJ (2004) Appropriating electronic billing systems: adaptive structuration theory analysis. *Human Systems Management* **23**(4), 235–243.
- STRAUSS A and CORBIN J (1990) *Basics of Qualitative Research: Grounded Theory Procedures and Techniques*. Sage Publication, London.
- THOMAS JC and BAKER SW (2008) Establishing an agile portfolio to align it investments with business needs. *Agile Conference*, pp 252–258, IEEE Computer Society, Toronto, ON.
- VIDGEN R and XIAOFENG W (2009) Coevolving systems and the organization of agile software development. *Information Systems Research* **20**(3), 355–376.
- WALLACE L and KEIL M (2004) Software projects risks and their effect on outcomes. *Communications of the ACM* **47**(4), 68–73.
- WEILL P (1991) The information technology payoff: implications for investment appraisal. *Australian Accounting Review* **1**(1), 2–11.
- WEILL P and BROADBENT M (1998) *Leveraging the New Infrastructure: How Market Leaders Capitalize on Information Technology*. Harvard Business School Press, Boston.
- WEILL P and OLSON MH (1989) Managing investment in information technology: mini case examples and implications. *MIS Quarterly* **13**(1), 3–18.
- YIN RK (2003) *Case Study Research: Design and Methods*. Sage Publications, London.
- ZIGURS I, DESANTIS G and BILLINGSLEY J (1991) Adoption patterns and attitudinal development in computer-supported meetings: an exploratory study with SAMM. *Journal of Management Information Systems* **7**(4), 51–70.

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